

Store-Level Time Series Forecasting of Restaurant Visitors Using ARIMAX

A Detailed Empirical Report Based on the Recruit Restaurant Visitor Forecasting Dataset

Abstract

This paper studies daily restaurant visitor forecasting using the Recruit Restaurant Visitor Forecasting dataset. The implemented comparison is deliberately focused on a weekly Seasonal Naive lag-7 benchmark versus a store-level ARIMAX model. The forecasting unit is `air_store_id` $\times$ `visit_date`, and the main evaluation metric is pooled observed store-date RMSLE on the final chronological holdout.

The final project follows the Kaggle prediction logic at the store-date level: the target is the number of visitors for each `air_store_id` on each date, rather than one aggregate visitor series for all restaurants combined. The central research question is whether a store-level ARIMAX model that uses weekly seasonality and economically meaningful exogenous variables can improve forecast accuracy relative to a simple weekly Seasonal Naive benchmark.

The theoretical framework treats restaurant demand as an operational decision problem under uncertainty. Daily demand is shaped by three mechanisms: persistence from recent store-specific demand, calendar-driven seasonality, and observable leading indicators such as reservations and holidays. The empirical pipeline constructs store-level daily series for 814 AIR stores, creates variables for holidays, weekends, Golden Week, and reservations, and evaluates forecasts on the final holdout window from 15 March 2017 to 22 April 2017. The baseline is Seasonal Naive with lag 7. The final model is ARIMAX(1, 1, 1)(1, 1, 1)[7] estimated separately by store with exogenous regressors.

Over 31,476 observed store-date holdout rows, ARIMAX reduces RMSLE from 0.9622 to 0.8122, a 15.6% improvement relative to the baseline, and outperforms the baseline for 79.1% of evaluated stores. A lightweight expanding-window validation fold provides additional support: when trained through 31 January 2017 and tested on 1–7 February 2017, ARIMAX reduces RMSLE from 0.9327 to 0.7317, a 21.6% improvement. The findings support the view that restaurant forecasting is more appropriate at the store level and should incorporate calendar information and reservation signals that reveal demand before the visit date, rather than relying only on repeated weekly history.

1. Introduction

Restaurant managers must make daily staffing, inventory, and capacity decisions before actual demand is fully observed. If the forecast is too low, the restaurant may face long waiting times, overworked staff, stock-outs, and lower customer satisfaction. If the forecast is too high, the restaurant may waste labor hours and perishable ingredients. Visitor forecasting is therefore not simply a technical prediction exercise; it is a practical economic problem about allocating limited resources under uncertainty.

This paper addresses the following question: can a store-level ARIMAX model using calendar and reservation information forecast daily restaurant visitors more accurately than a simple weekly Seasonal Naive benchmark? This question is intentionally precise. The project does not attempt to rank all possible time-series and machine-learning models. Instead, it tests a specific and defensible hypothesis: because restaurant demand is shaped by weekly routines, holidays, and advance reservations, a model that combines time-series dynamics with exogenous demand signals should outperform a benchmark that only repeats the previous week's observed pattern.

The empirical setting is the Recruit Restaurant Visitor Forecasting dataset. The dataset contains visitor counts, reservation records, store information, and calendar variables for restaurants in Japan. The final modelling notebook uses 814 AIR stores and evaluates 812 stores after applying the final modelling requirements. The final chronological holdout period runs from 15 March 2017 to 22 April 2017. The unit of analysis is a store-date row. This is crucial: the project is not forecasting one total market-level demand series, but daily demand for each individual store. This matches the real decision problem because a restaurant platform or manager needs a prediction for a specific restaurant on a specific day.

The paper makes four contributions within the scope of the implemented project. First, it aligns the empirical design with the actual prediction task by using store-level daily forecasting rather than aggregate forecasting. Second, it compares a transparent weekly benchmark with a final ARIMAX specification that incorporates holiday, weekend, Golden Week, and reservation features. Third, it treats pooled store-date RMSLE as the primary metric, while using the store-level win rate as supporting evidence that the improvement is not concentrated in a few restaurants. Fourth, it explicitly separates supported forecasting claims from extensions that were not implemented or were only lightly tested.

The main result is clear. On the pooled holdout set, the Seasonal Naive baseline achieves RMSLE of 0.9622, while ARIMAX achieves RMSLE of 0.8122. This is a 15.6% improvement. ARIMAX also performs better than the baseline for 79.1% of evaluated stores. The project also includes one lightweight expanding-window validation fold. In that earlier split, ARIMAX reduces RMSLE from 0.9327 to 0.7317, a 21.6% improvement. These results support the argument that store-level visitor demand contains information beyond repeated weekly seasonality. Weekly repetition is useful, but it is incomplete. Forecasting improves when the model also uses calendar variables and the reservation signal.

The rest of the report is organised as follows. Section 2 reviews the relevant forecasting literature and develops the theoretical framework. Section 3 describes the data, sample, variables, models, and evaluation strategy. Section 4 presents the main results. Section 5 discusses the economic interpretation, limitations, and future research directions. Section 6 concludes.

1.1 Store-level forecasting as the decision unit

The store-level design is not a cosmetic detail. It is central to the logic of the project. A restaurant platform may care about total demand across all restaurants, but operational decisions are made at the individual restaurant level. Staff are scheduled for a specific store, food is prepared in a specific kitchen, and service bottlenecks occur in a specific dining room. A forecast of total market demand could look accurate while still being useless for a manager if it hides large errors across individual restaurants.

This point also strengthens the use of pooled store-date RMSLE. The forecast task is not one series and one error per day; it is a panel of store-date forecasts. Pooled RMSLE treats each evaluated store-date row

as an operational prediction unit and summarizes performance across that panel. The store-level win rate then adds a second diagnostic by asking whether ARIMAX improves broadly across stores or whether the pooled result is driven by a small subset of restaurants.

The store-level design does introduce a trade-off. Estimating separate ARIMAX models by store means each model has fewer observations than a pooled global model would have. Some stores have shorter histories and noisier demand patterns. However, this trade-off is reasonable for the goal of the report. Separate store-level models allow each restaurant's history to drive its own forecast, which is consistent with the theoretical view that restaurants differ in clientele, location, cuisine, operating schedule, and reservation behavior.

2. Literature Review and Theoretical Framework

2.1 What the literature suggests about restaurant demand forecasting

The analysis begins with the behavior being modeled rather than with the model itself. Customer arrivals are not completely random. They are partly driven by habits, work schedules, leisure patterns, holidays, store popularity, and booking behavior. This means the time series has structure. The role of the empirical model is to represent enough of that structure to produce useful forecasts without making claims that exceed the data.

The classical time-series literature provides the first foundation. Box and Jenkins (1970) developed the ARIMA framework, where current values are modelled using autoregressive terms, differencing, and moving-average errors. For seasonal data, the framework extends naturally to include seasonal autoregressive and moving-average components. Restaurant visitor counts are a natural case for this logic because demand is often different on Monday, Friday, Saturday, and Sunday. A seven-day seasonal cycle is therefore grounded in the rhythm of the business rather than chosen arbitrarily.

The forecasting literature also emphasizes the importance of benchmarks. Hyndman and Athanasopoulos (2021) argue that more complex models should be judged against simple and transparent alternatives. For this project, the relevant benchmark is not a mean forecast but a weekly Seasonal Naive forecast. A restaurant that served a certain number of customers last Friday may reasonably expect the next Friday to be similar, especially when no other information is available. If ARIMAX cannot beat this benchmark, then its additional complexity is not justified.

Large-scale forecasting comparisons also caution against assuming that complexity automatically improves accuracy. Makridakis, Spiliotis, and Assimakopoulos (2018) show that simple statistical benchmarks remain difficult to beat in many forecasting settings. A related lesson from the forecast-accuracy literature is that evaluation must match the decision problem. Hyndman and Koehler (2006) warn that common percentage-error measures can behave poorly when actual values are close to zero, which matters in store-level restaurant data where some store-days have low counts.

A second relevant literature concerns dynamic regression and ARIMAX models. These models preserve the time-series structure of ARIMA while allowing external variables to enter the equation. For restaurant demand, holidays and reservations are examples of such information. A holiday may shift customer behaviour even if the previous week's pattern was different. A reservation made before the visit date may reveal future demand that has not yet appeared in the historical visitor series. The intervention-analysis

tradition of Box and Tiao (1975) is also relevant because public holidays and special calendar events can be treated as pre-known shocks to the demand process.

Taken together, the literature suggests three useful propositions: restaurant demand is dynamic, weekly seasonality is strong enough that a lag-7 benchmark is meaningful, and exogenous information can improve forecasts when it is known before the forecast origin. The gap addressed in this project is not the invention of a new forecasting model, but the disciplined implementation of a store-level forecasting design that links weekly seasonality, calendar structure, and reservations to a realistic chronological evaluation. This report therefore tests whether that theoretically motivated design can outperform a strong weekly benchmark in the implemented dataset.

2.2 Theoretical framework: demand under uncertainty

The theoretical framework views restaurant visitor forecasting as a decision problem under uncertainty. A restaurant manager must choose labour, ingredients, and table capacity before demand is realised. Forecasting therefore reduces uncertainty and supports better operational decisions. A better forecast does not create demand by itself, but it improves the match between expected demand and prepared capacity.

The framework has three channels. First, store-specific persistence matters: a restaurant that has recently attracted many customers may continue to do so because of location, reputation, cuisine type, and local routines. Second, calendar structure matters: workdays, weekends, public holidays, and extended holiday periods create different opportunity costs and leisure patterns. Third, advance demand revelation matters: reservations are observed before the visit occurs and therefore provide direct information about expected arrivals.

The empirical comparison follows directly from these channels. Seasonal Naive lag 7 captures the simplest form of calendar repetition: the same weekday last week. ARIMAX captures that weekly structure through seasonal terms and adds exogenous variables for the calendar and reservation channels. If ARIMAX improves forecast accuracy, the result supports the claim that the additional channels carry forecasting value.

This framework also limits the language of the study. The project is a forecasting exercise rather than a causal identification design. The holiday and reservation variables are used because they are available before the forecasted visit date and are theoretically informative about demand. The analysis therefore asks whether these variables improve prediction, not whether the project has estimated a causal effect of holidays or reservations on visitors.

2.3 Hypothesis derived from the framework

The theoretical framework implies the following hypothesis: a store-level ARIMAX model with calendar and reservation variables should forecast daily visitors more accurately than a Seasonal Naive lag-7 benchmark. The hypothesis is not that ARIMAX is universally best. It is more precise: within the implemented final project, ARIMAX should beat the weekly benchmark if exogenous demand signals contain useful information beyond last week's same-day demand.

Based on this framework, the empirical analysis evaluates three expectations:

- H1: weekly seasonality is strong enough that a lag-7 benchmark provides a meaningful forecasting standard;
- H2: calendar variables such as holidays and weekends provide predictive information beyond simple weekly repetition;
- H3: reservation counts act as leading demand indicators and improve store-level visitor forecasts when incorporated into a dynamic regression model.

Table 1: Theoretical channels and empirical representation

Theoretical channel	Operational interpretation	Empirical representation
Store-specific persistence	Demand is partly shaped by recent store-level history	ARIMA terms in ARIMAX
Weekly calendar structure	Customer behaviour differs by weekday and weekend	Seasonal period = 7 and <code>is_weekend</code>
Holiday shocks	Public holidays can shift eating-out behaviour	<code>is_holiday</code> and <code>is_golden_week</code>
Advance demand signal	Reservations reveal expected visitors before arrival	<code>log1p_reserve</code>

2.4 Hypotheses and expected empirical patterns

The theoretical framework leads to several expected patterns. First, the Seasonal Naive model should be a meaningful baseline rather than a trivial comparator. If weekly seasonality is genuinely present, last week’s same weekday should provide a reasonable first approximation of future demand. Second, ARIMAX should improve pooled RMSLE if the exogenous regressors contain predictive information beyond weekly repetition. Third, the improvement should appear not only in the pooled metric but also across a substantial share of stores. A pooled improvement without store-level consistency would be weaker because it could be caused by a few high-volume or unusually predictable restaurants.

These expectations make the evaluation criteria theoretically meaningful. Pooled RMSLE tests whether the final model lowers average log-scale error across the operational prediction rows. The store-level win rate tests whether the improvement is broadly distributed across the restaurant panel. The lightweight expanding-window fold tests whether the advantage appears in another chronological split, even though it is not a full rolling-origin validation design. Together, these criteria connect the empirical results back to the theory rather than treating evaluation as a separate technical step.

2.5 From theoretical channels to empirical variables

The target variable, daily visitors, represents realised restaurant demand. Because visitor counts are non-negative and right-skewed, the model uses the `log1p` transformation. This reduces the influence of very large visitor counts and aligns the estimation scale with RMSLE, where errors are measured on the log-transformed count scale.

The variable `is_weekend` represents predictable changes in customer opportunity cost and leisure time.

The variable `is_holiday` represents known public-holiday disruptions to ordinary weekly patterns. The variable `is_golden_week` represents a Japan-specific calendar period that is theoretically important because Golden Week is associated with travel and leisure. The variable `log1p_reserve` is the most direct example of advance demand revelation because it reflects information about future attendance that exists before the visit date.

This distinction between lagged demand and reservations is conceptually important. Lagged visitors summarise what happened in the past, while reservations describe commitments or intentions about a future date. Seasonal Naive can only repeat last week's same weekday and cannot react when future-date bookings change materially. ARIMAX can adjust the forecast when reservation information is already known at the forecast origin. This is the clearest theoretical reason why a dynamic regression model can outperform a weekly benchmark even when the weekly benchmark is already strong.

Table 2: Operationalisation of theoretical channels in the final project

Theoretical channel	Operational meaning	Empirical representation	Expected forecasting role
Store-specific persistence	Recent store demand contains information about future demand	AR and MA terms in ARIMAX	Captures dynamic dependence within each store
Weekly seasonality	Demand differs systematically across weekdays	Seasonal period 7 and Seasonal Naive lag 7	Captures repeated weekly rhythm
Calendar structure	Weekends and public holidays change behaviour	<code>is_weekend</code> , <code>is_holiday</code> , <code>is_golden_week</code>	Allows known date effects to shift forecasts
Advance demand revelation	Reservations reveal future demand before arrival	<code>log1p_reserve</code> from AIR + mapped HPG	Adds leading information unavailable to the naive benchmark
Scale heterogeneity	Restaurants differ greatly in visitor volume	<code>log1p_visitors</code> , RMSLE	Makes errors more comparable across stores

2.6 Why ARIMAX is more appropriate than ARIMA for this project

ARIMA would model each store's visitor series using its own past values, differencing, and moving-average errors. That is useful when the target series contains enough internal structure to predict itself. However, the theoretical framework says restaurant demand is not only internal. It is also affected by information that is known outside the target sequence, such as holidays and reservations. If these variables are excluded, their influence is pushed into the error term.

ARIMAX extends ARIMA by allowing exogenous regressors to enter the model. In this project, that extension is not optional; it is the empirical expression of the theory. The research question asks whether calendar and reservation information improve forecasting beyond weekly repetition. ARIMAX therefore

provides a direct way to evaluate the value of exogenous demand signals while maintaining a time-series structure.

This also explains why the Seasonal Naive benchmark is fair rather than weak. Seasonal Naive already captures one major theoretical channel: weekly repetition. It is a strong baseline for restaurant data because the same weekday from the previous week often contains meaningful information about operating schedules and customer routines. When ARIMAX beats this benchmark, the improvement can be interpreted as evidence that the additional channels in the theoretical framework carry forecasting value.

2.7 Predictive interpretation rather than causal interpretation

The final project is a forecasting study, not a causal identification study. The question is not whether the model has isolated a causal effect of holidays or reservations; the question is whether including known calendar information and reservation signals helps predict demand in a realistic forecasting setup. This predictive interpretation also explains why information timing is central: a variable is valid for forecasting only if it would be available at the forecast origin.

This distinction is especially important for reservations. A reservation variable is a leading indicator, but it is not necessarily a pure causal treatment. More reservations do not mechanically cause all visitor demand; rather, reservations reveal information about expected demand that is partially hidden from the historical visitor series. For a restaurant manager, this distinction is practically useful. The manager does not need the reservation coefficient to have a causal interpretation in order for reservations to improve forecasts. If reservation counts are observed before the visit date and are correlated with realised visitor demand, they can be valuable forecasting information.

The predictive framing also explains why the report emphasises forecast-origin discipline. Calendar variables such as weekend, public holiday, and Golden Week status are known in advance. Reservation variables are valid only to the extent that they are booked before the forecast origin. For that reason, the final notebook caps holdout reservation inputs at the training cutoff. This protects the evaluation from using reservations that would not have been known when the forecast was made. Without this discipline, the model could appear more accurate by accidentally using future information.

This narrower predictive claim is a strength rather than a weakness. The correct conclusion is not that holidays or reservations have been causally identified as drivers of restaurant demand. The correct conclusion is that a store-level ARIMAX model using calendar and reservation information improves out-of-sample forecast accuracy relative to a weekly Seasonal Naive benchmark in the implemented design. This keeps the empirical claim aligned with the evidence.

3. Data and Methodology

3.1 Data source and final sample

The study uses the Recruit Restaurant Visitor Forecasting dataset, which combines restaurant visits, reservations, store characteristics, and calendar information. The original Kaggle dataset contains reservation information from both the AIR and HPG platforms. The final project uses AIR reservations together with mapped HPG reservations as the operational reservation signal.

The final sample starts with 814 stores. Of these, 812 stores are evaluated in the final holdout and 2 are skipped because they do not meet the final modelling requirements. The final modelling notebook builds each store’s time series only over its observed date span rather than padding all stores with synthetic pre-history. Missing days inside a store’s observed span are filled as zero visitors to create the regular daily spacing required for time-series modelling.

The global visit range in the final notebook is 1 January 2016 to 22 April 2017, but the store-level series are built from each store’s own observed span. The final training period ends on 14 March 2017. The final holdout period runs from 15 March 2017 to 22 April 2017. The pooled final holdout contains 31,476 observed store-date rows.

Table 3: Final sample definition

Item	Final project value
Forecasting unit	<code>air_store_id × visit_date</code>
Stores total	814
Stores evaluated in final holdout	812
Stores skipped in final holdout	2
Global visit range	2016-01-01 to 2017-04-22
Final train end	2017-03-14
Final holdout window	2017-03-15 to 2017-04-22
Holdout observations	31,476 pooled observed store-date rows
ARIMAX fitting failures	0

3.2 Data cleaning and feature construction

ARIMAX needs a regular time index, so the final modelling notebook builds a daily frame for each store over that store’s observed date span. The dependent variable is daily visitors, and the transformed modelling target is `log1p_visitors`. The reservation feature is constructed by aggregating reservation visitors at the store-date level and applying `log1p`. The final reservation source combines AIR reservation records with mapped HPG reservation records.

The calendar variables include `is_holiday`, `is_weekend`, and `is_golden_week`. The rule for filling missing days is central to the store-level design: the final notebook builds each store only on its observed date span, then fills missing dates inside that span as zero visitors. This is an operational data-processing assumption for constructing a regular forecasting panel, not a causal claim about whether every missing raw-data date reflects closure, zero demand, or incomplete recording.

The final notebook also caps holdout reservation regressors at the forecast origin, 14 March 2017. This is an important leakage-control step because it prevents the model from using reservation records that would not have been known at the time the holdout forecast was generated.

Table 4: Core variables used in the final model

Variable	Definition in the project	Theoretical reason
visitors	Daily observed visitors for each air_store_id	Operational demand to be forecast
log1p_visitors	$\log(1 + \text{visitors})$	Stabilises scale and matches RMSLE
is_holiday	Public-holiday indicator	Captures abnormal calendar demand
is_weekend	Weekend indicator	Captures recurring non-working-day behaviour
is_golden_week	Golden Week indicator	Controls for a major Japanese holiday period
log1p_reserve	$\log(1 + \text{reserved visitors})$ by store-date	Leading indicator of future arrivals

3.3 Model 1: Seasonal Naive lag-7 baseline

The baseline model is Seasonal Naive with lag 7. For each store, the forecast for a date is based on the observed visitor count from the same weekday in the previous week. This is a strong baseline because it directly captures weekly seasonality, which is one of the clearest structures in restaurant demand. Its strength is simplicity; its weakness is that it assumes last week's pattern is sufficient.

3.4 Model 2: Store-level ARIMAX

The final model is ARIMAX(1, 1, 1)(1, 1, 1)[7] estimated separately for each store. The non-seasonal part models short-run store-specific dynamics. The seasonal part uses a seven-day period to capture weekly repetition. The exogenous regressors are is_holiday, is_golden_week, is_weekend, and log1p_reserve.

The order (1, 1, 1)(1, 1, 1)[7] is used as a theoretically motivated fixed specification. The differencing terms address trend and seasonal non-stationarity, the seasonal period of seven follows the dominant weekly pattern identified in the EDA, and the low-order AR and MA terms provide a parsimonious structure that remains computationally feasible across more than 800 stores. This choice is therefore justified by a balance of theory, runtime, stability, and interpretability rather than by exhaustive per-store tuning.

A fixed ARIMAX order is imposed across stores as a deliberate trade-off between local optimality and comparability. Allowing every store to have its own tuned ARIMA order might improve some individual-store forecasts, but it would also introduce a large model-selection layer, increase overfitting risk, and make the empirical comparison less transparent.

Table 5: Implemented model comparison

Model	Role	Information used	Main limitation
Seasonal Naive lag 7	Benchmark	Same weekday from previous week	Ignores reservations and holiday shocks
ARIMAX(1, 1, 1)(1, 1, 1)[7]	Final model	Store history, weekly seasonality, holidays, weekends, Golden Week, reservations	Fixed order and limited validation windows

3.5 Why a fixed ARIMAX order is defensible

The final model uses ARIMAX(1, 1, 1)(1, 1, 1)[7] for the store-level forecasts. A possible criticism is that different stores might prefer different ARIMA orders. That criticism is reasonable, but it does not invalidate the implemented design. The project chooses a fixed order as a trade-off between local optimality, comparability, runtime, and reproducibility. If every store had a separately tuned order, the project would introduce a second modelling layer: model selection for hundreds of individual stores. That could improve some store-level forecasts, but it would also increase complexity and overfitting risk.

A fixed order keeps the empirical question clean. The report is not asking whether the best possible ARIMA order can be found for every restaurant. It asks whether one coherent store-level dynamic regression specification, motivated by weekly seasonality and exogenous demand signals, can improve over a strong weekly benchmark. The common order makes stores comparable and allows the performance difference to be attributed to the overall modelling framework rather than to a complicated search procedure.

The fixed-order choice is therefore a deliberate design decision rather than an ignored weakness. It is a limitation because some stores may have different dynamics. However, it is also a strength because it creates a transparent empirical test. The model has a known structure, the benchmark has a known structure, and the evaluation compares them over the same store-date holdout rows. That simplicity makes the empirical comparison easier to understand and easier to reproduce.

3.6 Forecast-origin discipline and leakage control

A central requirement in time-series forecasting is that the model must not use information that would be unavailable at the time the forecast is made. Reservation data require special care because a reservation record contains both a reservation time and a visit time. The final notebook therefore caps holdout reservation regressors at 14 March 2017 for the final holdout. This means the model does not use reservation information created after the training cutoff when forecasting the holdout period.

This leakage-control step also strengthens the credibility of the ARIMAX improvement. Because `log1p_reserve` is a leading indicator, it could artificially inflate performance if constructed incorrectly. By capping reservation inputs at the forecast origin, the project demonstrates that the exogenous variable is being used as information available before the forecast, not as information retrospectively observed after the outcome. This makes the comparison with Seasonal Naive fairer and makes the reported improvement

more defensible.

The treatment of missing dates is another important methodological issue. ARIMAX and the lag-7 baseline require a regular daily time index. The final project avoids synthetic pre-history padding and models each store only over its observed date span. Within an observed span, missing days are filled as zero visitors. This is a pragmatic modelling assumption that allows the time index to remain regular. It should not be overinterpreted as a causal statement about whether every missing raw record is definitely a closed day or a true zero-demand day.

3.7 Reproducibility and final project organization

The final notebook reads processed inputs from `notebooks/processed_data/` and writes corrected outputs to the `output/` folder. The final results use the `_v2` output files, which separates corrected results from earlier experiments. The repository also includes a requirements file listing the core Python packages needed to run the notebook pipeline.

The project organisation supports reproducibility by presenting the workflow in a clear order: overview, final results, final modelling notebook, cleaning and feature engineering, and EDA. The final submission files include the notebook, the results summary, and saved output files. This structure makes the empirical evidence traceable: a reader can move from the reported numbers back to the notebook and from the notebook back to the prepared data.

This naming discipline is especially useful because forecasting projects often go through several versions as leakage, padding, or evaluation issues are fixed. By explicitly using only the corrected `_v2` outputs for the final claim, the project reduces ambiguity and strengthens reproducibility. The requirements file reinforces the same point by showing that the final project is organised as a reproducible notebook pipeline rather than a one-off calculation.

3.8 Evaluation strategy and metrics

Because the data are time series, the evaluation uses chronological splitting rather than random sampling. The main reported metric is pooled RMSLE over all observed store-date rows in the final holdout set. This pooled metric is the primary basis for model comparison because the forecasting task is defined at the store-date level and the holdout contains many restaurants observed over the same final test window.

MAPE is reported as a secondary metric. It is intuitive because it is expressed as a percentage, but it becomes unstable when actual visitor counts are small. For that reason, the main judgement is based on RMSLE and the store-level win rate. The final notebook also includes one lightweight expanding-window validation fold with training ending on 31 January 2017 and a test window covering 1 February to 7 February 2017. This fold is interpreted as a validation check, not as full rolling-origin cross-validation.

Chronological validation is necessary in this project because the order of observations carries information. A random train–test split would be inappropriate because it could place future observations in the training set and earlier observations in the test set. That would create an evaluation environment unlike the real forecasting problem, where the future is not known when the prediction is made. The final holdout therefore comes after the training period, and the lightweight validation fold also respects time order.

RMSLE is particularly suitable because visitor counts are non-negative and heterogeneous across stores. A raw error of 10 visitors has a different meaning for a small restaurant that usually serves 15 visitors than for a large restaurant that usually serves 120 visitors. Log-scale errors reduce the dominance of large stores and make proportional differences more visible. The use of $\log_{1p}$ also handles zero values and keeps the modelling scale aligned with the main evaluation metric.

3.9 Methodological safeguards

The credibility of the project depends on four safeguards: chronological evaluation, a meaningful seasonal benchmark, a store-level forecasting unit, and reservation features capped at the training cutoff. Together, these safeguards make the final metrics interpretable and ensure that the supported conclusion remains aligned with the implemented design.

These safeguards matter because final metrics are persuasive only when the design that produced them is defensible. A strong benchmark prevents easy wins against weak comparators. Chronological evaluation prevents future information from contaminating the test period. The store-level forecasting unit keeps the empirical design aligned with the operational decision problem. Forecast-origin discipline prevents the reservation signal from becoming a form of hidden leakage. Together, these choices support a coherent time-series methodology rather than a loosely assembled predictive exercise.

3.10 Implementation pipeline summary

The final project can also be understood as a reproducible sequence of transformations from raw platform data to evaluated forecasts. First, the cleaning pipeline standardises dates, merges store identifiers, and aggregates reservations to the store-date level. Second, the feature-engineering pipeline creates a regular daily panel within each store's observed span, fills missing dates with zero visitors, and constructs the calendar and reservation regressors used in the final notebook. Third, the modelling stage fits the Seasonal Naive lag-7 benchmark and the fixed-order ARIMAX specification separately by store. Fourth, the evaluation stage pools observed store-date predictions, computes RMSLE as the primary metric, and summarises store-level win rates and lightweight validation performance.

This pipeline matters because each stage corresponds to a specific forecasting principle. Regular daily spacing is needed for lag-7 comparisons and seasonal ARIMA estimation. Forecast-origin discipline is needed to keep reservation features valid. Separate store-level estimation is needed to preserve the operational decision unit. Chronological holdout evaluation is needed to keep the test environment realistic. The final notebook therefore functions not only as code, but also as the operational definition of the empirical design described in this report.

Table 6: End-to-end forecasting pipeline in the final project

Stage	Implemented action	Forecasting purpose
Raw data integration	Merge visitors, store metadata, calendar variables, AIR reservations, and mapped HPG reservations	Bring all predetermined demand signals into one store-date pipeline
Panel construction	Build a daily frame within each store's observed span and fill internal missing dates as zero visitors	Create the regular spacing required for lag-7 forecasting and ARIMAX estimation
Feature engineering	Construct <code>log1p_visitors</code> , <code>is_holiday</code> , <code>is_weekend</code> , <code>is_golden_week</code> , and <code>log1p_reserve</code>	Operationalise the theoretical demand channels
Model estimation	Fit Seasonal Naive lag 7 and ARIMAX(1, 1, 1)(1, 1, 1)[7] separately by store	Compare a strong weekly baseline with a store-level dynamic regression model
Chronological evaluation	Score the final holdout and one expanding-window fold using pooled RMSLE and store-level win rates	Keep the evidence aligned with realistic forecasting and store-level decisions

4. Results

4.1 Main pooled forecast accuracy

The main result is that ARIMAX outperforms the Seasonal Naive baseline on the pooled final holdout set, which is the primary evaluation criterion in the report. Baseline pooled RMSLE is 0.9622. ARIMAX pooled RMSLE is 0.8122. This is a 15.6% improvement relative to the benchmark.

The reduction is operationally meaningful because forecast improvements at the store-day level can plausibly support better staffing, ingredient preparation, and inventory planning. MAPE is reported only as a secondary diagnostic and also improves, falling from 131.82% for the baseline to 86.99% for ARIMAX. Because MAPE is sensitive to low visitor counts, the headline result should still be interpreted through RMSLE rather than percentage errors.

Figure 1 makes the main result visually immediate. The gap between the two bars is not tiny relative to the scale of the metric, which reinforces the claim that the improvement is meaningful rather than cosmetic. Since the benchmark is already strong, reducing pooled RMSLE by this margin suggests that weekly repetition alone leaves predictive information unused. Calendar and reservation signals appear to provide that additional information.

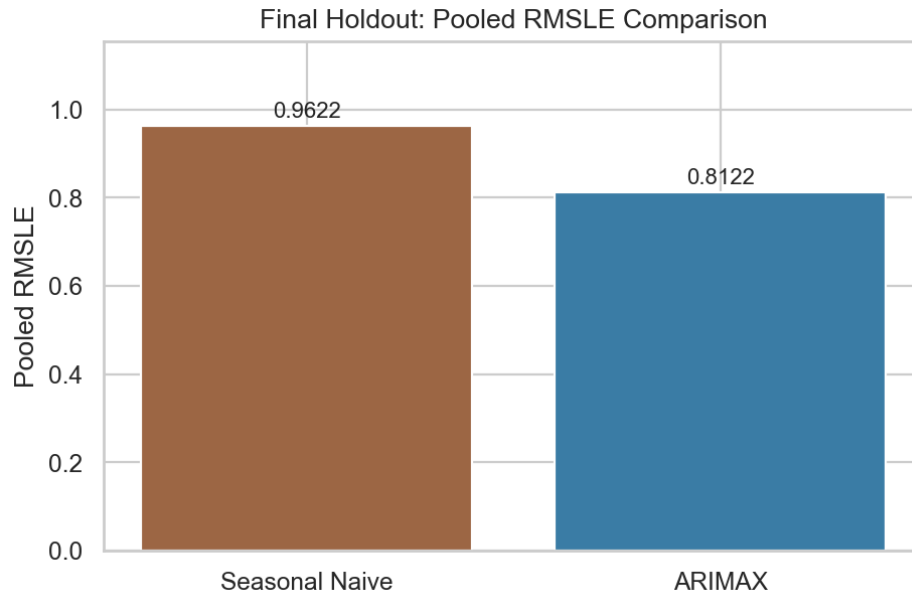


Figure 1: Final holdout pooled RMSLE comparison between Seasonal Naive and ARIMAX.

Table 7: Main pooled forecast metrics on the final holdout

Model	RMSLE	RMSE _{log}	MAPE (%)
Seasonal Naive lag 7	0.9622	0.9622	131.82
ARIMAX	0.8122	0.8177	86.99

4.2 Store-level consistency of the result

The store-level result supports the pooled result. ARIMAX beats the baseline for 79.1% of evaluated stores. This means the improvement is not concentrated in a small number of restaurants. Instead, the final model improves accuracy for a clear majority of stores in the sample.

Figure 2 adds an important diagnostic to this point. The mass of the distribution lies to the right of zero, which indicates that positive improvement is common across stores rather than being concentrated in a few outliers. This does not mean that ARIMAX wins everywhere, but it does mean that the pooled result is consistent with the store-level evidence.

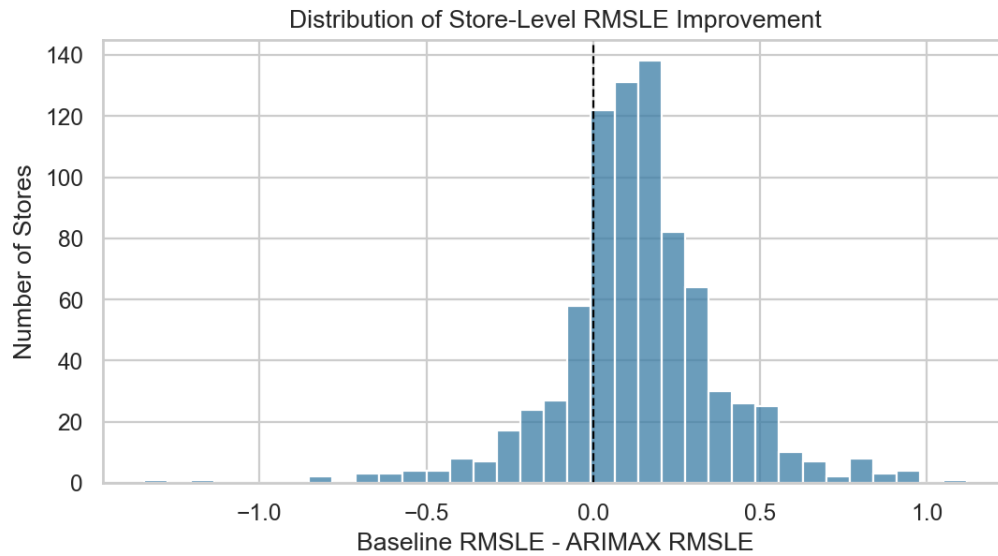


Figure 2: Distribution of store-level RMSLE improvement, measured as baseline RMSLE minus ARIMAX RMSLE. Values above zero indicate that ARIMAX outperformed the benchmark.

Table 8: Store-level summary of final holdout performance

Statistic	Value
Stores total	814
Stores evaluated	812
Stores skipped	2
Holdout observations	31,476
ARIMAX better than baseline	79.1% of evaluated stores
Improvement vs baseline	15.6% RMSLE reduction
ARIMAX fitting failures	0

4.3 Lightweight expanding-window validation result

The final project includes one lightweight expanding-window validation fold before the final holdout. The fold trains through 31 January 2017 and tests on 1 February to 7 February 2017. In this validation check, 810 stores are evaluated and the pooled validation set contains 5,670 store-date rows.

The result is consistent with the final holdout. The Seasonal Naive baseline achieves RMSLE of 0.9327, while ARIMAX achieves RMSLE of 0.7317. This corresponds to a 21.6% improvement. ARIMAX beats the baseline for 69.4% of evaluated stores in this fold.

This result should be interpreted carefully. This is one lightweight expanding-window fold, not a full multi-fold rolling-origin validation design. Therefore, it supports the robustness of the final result but does not prove stability over all possible forecast origins.

Even so, the fold is methodologically useful because it shows that the ARIMAX advantage is not confined to one arbitrarily chosen final window. In forecasting work, a single final holdout can always raise the question of whether the result depends too heavily on the exact test period. This earlier split does not fully

answer that concern, but it makes the final claim more credible.

Table 9: Lightweight expanding-window validation fold

Validation item	Value
Train end	2017-01-31
Test window	2017-02-01 to 2017-02-07
Stores evaluated	810
Pooled rows	5,670
Seasonal Naive RMSLE	0.9327
ARIMAX RMSLE	0.7317
Improvement vs baseline	21.6%
ARIMAX better than baseline	69.4% of evaluated stores

4.4 Why ARIMAX improves over the baseline

The improvement can be explained through the theoretical framework. Seasonal Naive captures only one source of predictability: weekly repetition. It cannot respond to changing conditions such as public holidays, shifts in reservation demand, or store-specific dynamics that are not well represented by last week's same weekday alone.

ARIMAX improves because it combines three types of information. First, it uses autoregressive and moving-average components to model store-specific persistence. Second, it includes seasonal structure with a seven-day period. Third, it adds calendar and reservation features. The model therefore has a richer but still interpretable structure. Its improvement comes from variables that have clear economic meaning rather than from opaque black-box behaviour.

Figure 3 illustrates this interpretation at the store level. The purpose of the plot is not to prove that ARIMAX dominates for every single date, but to show how the model can adapt differently from the weekly benchmark when actual demand departs from last week's pattern. This is where the reservation and calendar channels are most useful conceptually.

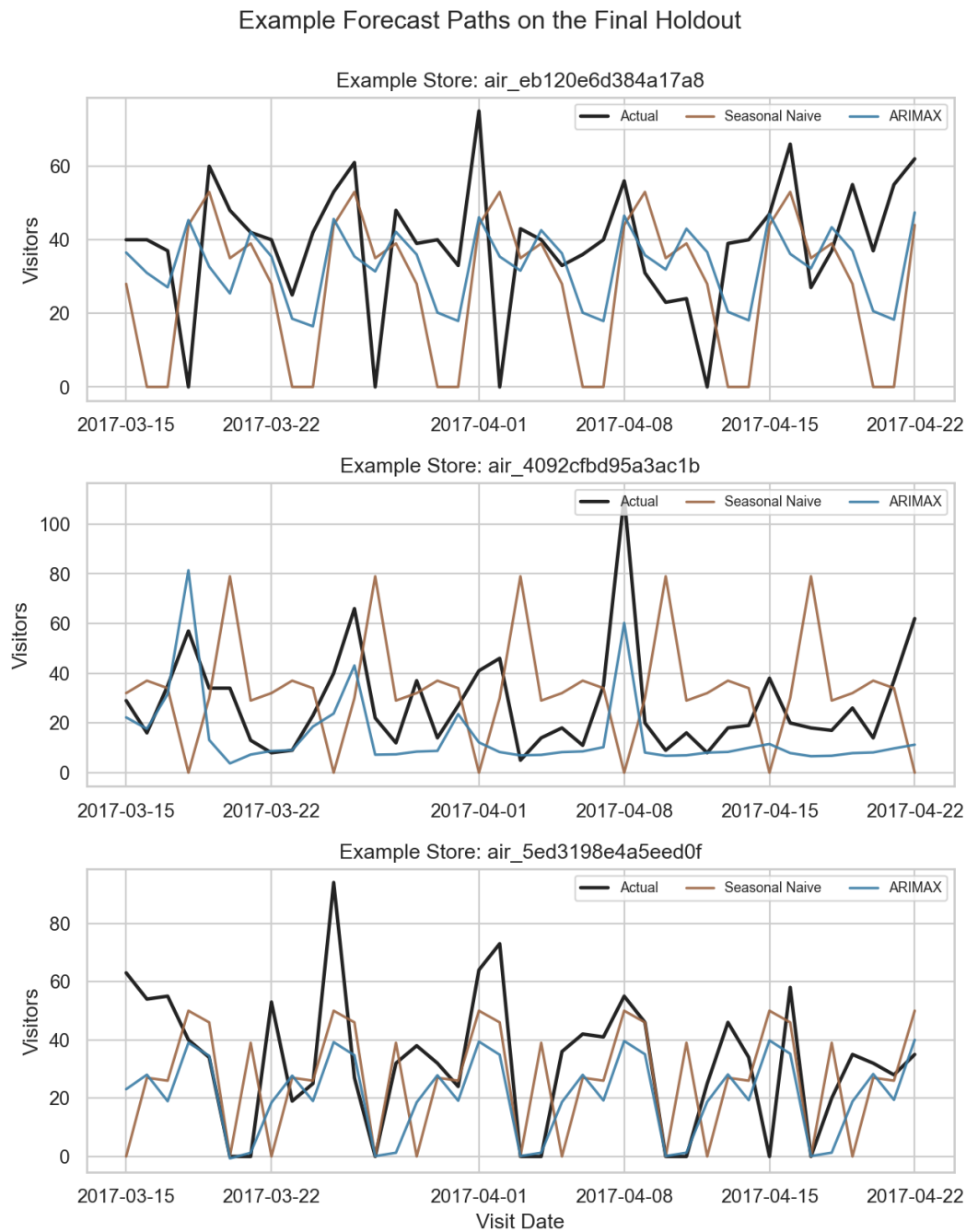


Figure 3: Example forecast paths for three stores on the final holdout window.

4.5 What the results do not prove

The results must be interpreted within the implemented design. They do not prove that ARIMAX is the best possible forecasting model across all alternatives, nor that one final holdout plus one lightweight validation fold establishes universal stability across all forecast origins. They also do not directly prove Golden Week forecasting performance, because the final holdout does not cover the main Golden Week dates.

4.6 Interpreting the magnitude of the improvement

The final holdout result shows a reduction in pooled RMSLE from 0.9622 for Seasonal Naive to 0.8122 for ARIMAX. The relative improvement is 15.6%. In a forecasting project, this is a meaningful result because the benchmark is not weak. Seasonal Naive already incorporates the most obvious pattern in restaurant data: weekly repetition. Beating such a benchmark means ARIMAX is not merely improving over a naive average or an unrealistic comparator.

The magnitude should still be interpreted carefully. A 15.6% reduction in RMSLE does not mean the forecast is perfect, nor does it mean all individual store-day forecasts are accurate. Restaurant demand remains noisy. Weather, local events, closures, promotions, and random customer behaviour can all affect visitors in ways not captured by the implemented variables. The result means that, on average over the evaluated store-date rows, ARIMAX produces smaller log-scale errors than the benchmark.

The store-level win rate provides additional interpretation. ARIMAX performs better than the baseline for 79.1% of evaluated stores in the final holdout. This matters because it shows that the pooled improvement is not narrowly concentrated. If ARIMAX had improved pooled RMSLE but won for only a small share of stores, one might suspect that the result was driven by a few high-volume restaurants. Instead, the win rate suggests that the model's advantage is broad across the store panel.

The lightweight expanding-window fold sharpens that interpretation. In the earlier fold, ARIMAX improves RMSLE from 0.9327 to 0.7317, a 21.6% improvement, and wins for 69.4% of evaluated stores. The win rate is lower than in the final holdout, but it still favours ARIMAX clearly. This pattern is useful because it shows that the model's advantage is not unique to the final March–April window. At the same time, one fold is not enough to prove stability across all seasons, so the validation result should be read as supporting evidence rather than decisive proof of generality.

4.7 Residual error interpretation and the role of MAPE

MAPE remains high even after the ARIMAX improvement: 131.82% for the baseline and 86.99% for the final model. At first glance, those numbers can look alarming. However, they must be interpreted in the context of store-level restaurant data, where some stores and days have very low visitor counts. When actual demand is close to zero, even a small absolute error can become a very large percentage error. This mechanical feature of MAPE is the main reason it is not used as the headline criterion in the report.

RMSLE is more appropriate here because it better reflects the scale heterogeneity of the problem. The metric reduces the influence of very large stores, accommodates zeros through the $\log_{1p}$ transformation, and corresponds more closely to the idea that proportionate misspecification matters more than raw absolute deviation. A forecast error of eight visitors does not have the same meaning for all stores, and RMSLE handles that better than a simple percentage measure.

For that reason, MAPE is retained mainly as a secondary descriptive diagnostic. Its directional movement is still useful: ARIMAX improves MAPE in the same direction as RMSLE, which is reassuring. But the substantive judgement of the paper rests on pooled RMSLE and store-level consistency, not on percentage errors that are known to become unstable in sparse count settings. This is a methodological choice grounded in the structure of the data rather than an after-the-fact attempt to downplay an inconvenient statistic.

5. Discussion

5.1 Practical implications and interpretation of Golden Week

The practical value of the model comes from reducing uncertainty before operational decisions are made. Restaurant managers must decide staffing, food preparation, and table allocation before the actual number of visitors is observed. In this setting, reservation information is particularly valuable because it is available before the visit date and directly relates to expected arrivals. Calendar variables are also useful because they are known in advance. By combining store-specific history, weekly seasonality, calendar conditions, and reservations, the ARIMAX model is more aligned with operational planning than a benchmark that only repeats last week's demand.

Improved visitor forecasts can affect several decisions. Underprediction can lead to understaffing, slower service, and stock-outs, while overprediction can lead to excess labour cost and food waste. A model that reduces log-scale forecast error can therefore have practical value even if it does not eliminate uncertainty. The store-level nature of the project is especially important here. Platform-level forecasts may be useful for high-level planning, but managers need store-specific daily predictions. The store-level win rate suggests that the ARIMAX framework is useful across many individual restaurants rather than only improving a market-level average.

Golden Week should be interpreted carefully. It is theoretically relevant because it is a major holiday period in Japan and can affect restaurant demand through travel, leisure, and family behaviour. Including `is_golden_week` is therefore reasonable as a calendar feature. However, the final holdout period ends on 22 April 2017, before the main Golden Week dates. The project therefore treats Golden Week as a known calendar control rather than as a directly validated holiday-stress-test result.

5.2 Limitations, future extensions, and theoretical coherence

The study has several limitations, all of which follow from its deliberately focused empirical design. First, the project compares only two implemented models: Seasonal Naive lag 7 and store-level ARIMAX. It does not run a broad model tournament involving Prophet, ETS, gradient boosting, random forests, LSTM, temporal fusion transformers, or hierarchical global models. The conclusion should therefore remain aligned with the implemented design: ARIMAX outperforms the implemented Seasonal Naive benchmark.

Second, validation remains limited. The use of a final chronological holdout and one lightweight expanding-window fold is appropriate for time-series forecasting and avoids random-split leakage, but it does not establish performance across all possible forecast origins. Future work should apply additional rolling-origin or expanding-window splits to test whether the ARIMAX advantage remains stable across different demand regimes and seasons.

Third, the construction of a regular daily store panel is a modelling assumption rather than an economic claim. The notebook builds a complete daily frame within each store's observed date span and fills missing days as zero visitors in order to preserve the regular spacing required for the lag-7 benchmark and ARIMAX. Future work could improve this step by incorporating explicit store closure calendars or richer indicators of operating status.

Fourth, the fixed ARIMAX order improves comparability and reproducibility, but it may not be individually optimal for every store. Future work could compare the fixed-order design against automated per-store order selection, hierarchical forecasting, pooled panel models, or global forecasting models that borrow strength across similar restaurants. Another useful extension would be to evaluate forecast performance through explicit business costs, such as staffing shortages, excess labour, or food waste, rather than relying only on statistical error metrics.

Despite these limitations, the empirical design remains theoretically coherent. Because demand is store-specific, the project forecasts at the store level. Because demand has weekly structure, the benchmark is Seasonal Naive lag 7 and the ARIMAX seasonal period is seven. Because demand responds to calendar conditions, the model includes holiday, weekend, and Golden Week indicators. Because reservations reveal demand in advance, `log1p_reserve` enters as an exogenous variable. Because the outcome is a non-negative count with skewness, the model uses `log1p_visitors` and evaluates primarily with pooled RMSLE. Each major technical choice therefore follows from the economic logic of the problem, which strengthens the credibility of the final result.

5.3 From forecast accuracy to operational value

The report evaluates forecasting quality statistically, but the practical motivation is operational. A restaurant manager does not observe RMSLE directly in daily work. What the manager observes are staffing mismatches, food waste, stock-outs, waiting times, and service congestion. A model that reduces forecast error is valuable because it changes those downstream decisions. Better anticipation of a busy Friday can mean one more staff member on shift, more ingredients prepared in advance, or a more realistic expectation of table turnover.

This point is especially relevant for reservation-rich environments. Reservations are not only data points for a statistical model; they are partial commitments from customers. When those commitments are combined with weekly history and calendar effects, the resulting forecast becomes closer to the information set that a manager actually has before service begins. That is why the reservation variable matters conceptually even though the report does not interpret its coefficient causally. The economic logic is that known future demand signals should improve planning when they are incorporated in a disciplined forecasting pipeline.

The store-level design strengthens this operational interpretation. Even if an aggregate forecast across all restaurants were accurate, it could still be useless for decision-making if overprediction in one store cancelled underprediction in another. Labour is scheduled by store, inventory is prepared by kitchen, and service bottlenecks arise locally. The gain from ARIMAX is therefore most meaningful when interpreted as an improvement in store-date planning rather than as an abstract market-level accuracy gain.

6. Conclusion

This report presents a revised version of the restaurant visitor forecasting project that matches the latest implemented notebook. The final empirical task is daily store-level forecasting for `air_store_id × visit_date` observations. The implemented comparison is between a Seasonal Naive lag-7 benchmark and an $\text{ARIMAX}(1, 1, 1)(1, 1, 1)[7]$ model with exogenous variables for holidays, weekends, Golden Week, and reservations.

The results show that ARIMAX performs better than the baseline on the final holdout window under the primary pooled RMSLE criterion. It reduces pooled RMSLE from 0.9622 to 0.8122, corresponding to a 15.6% improvement. It also beats the baseline for 79.1% of evaluated stores, which supports the view that the improvement is broad across the sample rather than concentrated in a small subset of restaurants. A lightweight expanding-window validation fold gives additional support: ARIMAX reduces RMSLE from 0.9327 to 0.7317 in the February validation window, a 21.6% improvement.

The conclusion is deliberately precise. The project does not claim to compare every possible forecasting model. It does not claim direct out-of-sample validation for Golden Week. It does not claim that one final holdout plus one lightweight validation fold proves stability across all future periods. What it does show clearly is that, in the implemented store-level design, ARIMAX adds forecasting value relative to a strong weekly Seasonal Naive benchmark.

This precision matters because empirical credibility depends on the match between evidence and claim. The project supports a specific conclusion: in the final store-level holdout, a dynamic regression model using calendar and reservation information improves over a weekly seasonal benchmark. It does not require broader language to be convincing. The strength of the report is that the research question, theory, variables, model, and evidence all point in the same direction.

The broader implication is that restaurant forecasting should be built around the actual decision unit. Managers and platforms need store-specific daily forecasts, not only aggregate demand summaries. Within the scope of this project, ARIMAX provides a logical and empirically supported way to combine store-level history, weekly seasonality, calendar information, and reservation signals.

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